

HAEMIN PARK

Industrial Engineering and Management Science
Northwestern University

Developing **efficient** and **robust** machine learning methods, both theoretically grounded and practically applicable. Particularly focused on **distributed** training, **continual** learning, memory/communication-**efficient** training, defect detection.

Education

Seoul Science High School, Seoul : (2013.3-2016.2)

Seoul National University, Seoul : (2016.3 - 2022.8)

Bachelor of Engineering (overall GPA: 3.87/4.3, 142 credits, Double Major in Industrial Engineering & Mathematics)

Northwestern University, Evanston : (2022.9 - **2027.06** (expected))

Industrial Engineering and Management Science ph.D (GPA: 3.88/4.00 (18units))

Research & Project Experience

Parameter&Communication-Efficient Federated Learning (2023-2025, **Under Review on TMLR**)

- Analyzed the low-rank nature of federated learning theoretically, and suggest parameter-efficient federated learning algorithm using low-rank updates.

Per-Class-Gradient Momentum for Class-Imbalance Problem & Efficient Online Hyper-Parameter Tuning in Federated Learning (2024-2025, **2026 ICML**)

- Developed a per-class gradient momentum technique, applying momentum and normalization to reduce directional noise, mitigate imbalance, and boost performance on imbalanced datasets.
- Developed the first locally adaptive hyper-parameter method for federated learning using a hierarchical partitioning strategy.

Privacy-preserving and NonIID-aware Federated Deep Embedded Clustering (2025-2026, **Under Review on AAAI**)

- Developed a federated deep embedded clustering method leveraging sample stability and reclustering based on aggregation noise inherent in federated learning to handle data privacy constraints.

Federated Continual Learning using Decoupled LoRA (2025-2026, **Works In Progress**)

- Developed **Decoupled LoRA with SAM (Sharpness-Aware Minimization)** for continual learning by separating update magnitude and direction, improving adaptation to new tasks while reducing catastrophic forgetting through flatter and more stable low-rank update geometry.

The Advantage of Low-Precision and Low-Rank Training in Distributed Systems (2025-2026, **Works In Progress**)

- Established that resource-efficient (low-precision/low-rank) training outperforms full-resource methods in large distributed settings, providing a formal analysis of their scaling behavior.

Industrial Experience

Intel Corporation - AI Research Engineering Intern (2026.06 - 2026.09)

- Researched diffusion-based synthetic defect generation for industrial manufacturing images, addressing the scarcity of labeled anomaly samples and rare geometric pattern of chip manufacturing images.
- Developed a rare-event-driven anomaly generation framework that models defects as localized deviations from normal images, with the goal of improving downstream defect detection.

Intel Corporation - AI Research Engineering Intern (2025.06 - 2025.09)

- Developed an evaluation pipeline to identify Vision Transformer architectures balancing inference latency, energy efficiency, and class-imbalance robustness.
- Proposed adaptive model selection criteria for practical deployment in manufacturing-quality visual inspection.

Common Computer : D-App & AI Developer (2020.10 - 2021.04)

- Blockchain, Cloud Computing and Artificial Intelligence Start-up in South Korea.
- Developed new blockchain SNS application in Japan and worked on the projects to deploy the cloud-computing resource allocation and online ad-allocation problem

Related Skills

Programming Skills

- Professional: Python, PyTorch, Anaconda, Git, Docker, JAVA, JavaScript, React, Firebase
- High-Level : C, R

Quantitative Skills

- Linear Algebra, Measure Theory, Financial Mathematics, Nonlinear Optimization, Predictive Analysis (Regression, Classification), Non-parametric Statistics, Time-Series Analysis, Stochastic Processes

Machine Learning and AI

- Federated Learning, Nonlinear Optimization, Privacy-Preserving Algorithms, Compression and Sparsification, Hyper-Parameter Optimization, Image classification, LLM Fine-Tuning, Transformers for Time-Series, Large-Scale Image Dataset